卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0617 30-B003701S-T0617

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 06 卷 第 17 册
Structure SPECIALITY VOLUME BOOK

卷册名称 直流场避雷线塔桩位图
Volume name layout of piles for tower of guard cable

批 准 APPROVED BY	仇子品
审 核 REVIEWBD BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	庞金来
图 纸 DRAWING 2 张 NOS.	打印件 PRINTED COPIES 1 本 NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0617-001	灌注桩设计说明 specification for cast-in-place pile	A	1		
1	30-B003701S-T0617-002	直流场避雷线塔桩位图 layout of piles for tower of guard cable	A	1		

注:

机械成孔灌注桩设计说明

specification for cast-in-place pile

1.概述

1.1 场地±0.000相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。

1.2本工程基础设计依据《巴基斯坦默蒂亚里±660KV直流输电工程拉合尔±660KV换流站施工图设计阶段岩土工程勘察报告（详勘）》(B003701S-G01-01)进行。

勘察报告显示：该工程场地12米以浅深度范围可能发生液化。

1.3本工程机械成孔灌注桩采用端承摩擦桩，桩端持力层为②3层②4 层粉细砂层。

1.4 参考国家建筑标准设计图集《钢筋混凝土灌注桩》图集号：10SG813；

2.成孔

2.1桩基施工过程中，应在成孔后及钢筋笼安放好后两次清理孔底沉渣，清孔后沉渣厚度不得大于100，符合规定后半小时内必须立即浇灌混凝土。否则应按重新清孔至符合要求。

2.3成孔达到设计深度后，孔口应予以保护，按有关规定验收，并做好记录。

2.4在施工过程中，应均匀随机抽查桩的孔径，数量应大于总桩数的10%。

3.钢筋笼制作及安装

3.1机械成孔灌注桩的主筋，螺旋箍筋和加劲箍筋等焊接成钢筋笼，若分段制作，钢筋笼连接时应采用焊接，纵横钢筋交接处均应焊牢，焊条采用E43及E50型。

3.2主筋和箍筋均不做弯钩，最低一道箍筋应对准主筋底端焊牢。主筋在同一截面焊接根数不得大于50%。

3.3钢筋笼外侧需设混凝土垫块或其他有效排施，以确保钢筋笼主筋保护层厚度为75mm。

3.4在施工过程中，钢筋笼搬运时应采取适当排施防止扭转，弯曲，然后对准孔位吊直轻稳缓慢下沉，避免碰撞孔壁，钢筋笼下放到设计位置后应固定。

3.5钢筋笼的制作允许偏差

主筋间距	±10mm	箍筋间距	±20mm
钢筋笼直径	±10mm	钢筋笼长度	±100mm

3.6分段制作的钢筋笼，其接头宜采用焊接（钢筋直径大于20mm），并应遵守国家现行标准《钢筋机械连接通用技术规程》JGJ 107、《钢筋焊接及验收规程》JGJ 18和《混凝土结构工程施工质量验收规范》GB 50204的规定。

3.7加劲箍设置参见图集(10SG813)。

3.8导管接头处外径应比钢筋笼的内径小100mm以上。

4.混凝土的充盈系数为>1.05。

5.施工允许偏差

5.1桩径D容许偏差为+50(不允许负偏差)。

5.2桩垂直度容许偏差为1/100。

5.3桩中心位置容许偏差：

（1）单排桩，条形桩基沿垂直轴线方向和群桩基础边桩的偏差为D/6且不大于100，但相邻两桩不能偏往同一方向。

（2）条形桩基沿轴线方向和群桩基础中间桩的偏差为D/4且不大于150，但相邻两桩不能偏往同一方向。

（3）柱下单桩的中心位置允许偏差为D/6且不大于100（D为桩径）。

6.施工时应进行监测，监测项目包括孔隙水压力观测、桩顶位移观测、土体位移观测等，根据监测结果情况，控制合理成桩速率及顺序，必要时采取措施，确保工程桩施工质量。

7.成桩后应休止一段时间再进行基坑开挖，具体休止时间及开挖日期应根据有关规范要求及超孔隙水压力消散情况确定。

8.成桩顺序原则：

8.1根据桩的密集程度，成桩可采用自中间向两个方向对称进行、自中间向四周进行、自一侧向单一方向进行。

8.2根据基础的设计标高，宜先深后浅。

8.3根据桩的规格，宜先大后小，先长后短。

8.4当一侧毗邻建筑物时，由毗邻建筑物处向另一方向施打。

8.5对于不同类型的桩，如管桩、灌注桩与搅拌桩、碎石桩一起施工时，宜根据实际情况，与设计共同协商桩的施工顺序。

9. 基坑开挖前应对边坡稳定、降水排施、挖土方案、运土路线、堆土位置等编制施工方案，坑内挖土应分层进行，高差不宜超过1米，随挖随运，不得堆置在坑边影响桩身稳定及防止桩的位移和倾斜。

10. 成桩过程中如发现地质条件与勘测资料不符，或其它意外情况，请及时与设计院联系。

11. 桩与承台的连接构造参见图集(10SG813)34页 桩顶与承台连接构造，桩内主筋伸入承台长度为40d。

12、质量检测与验收

12.1 质量标准除满足上述条款外，尚应遵循以下现行规范中的有关条款执行：

《Design, Manufacture, and Installation of Concrrete Piles》 ACI-543R-12

《建筑桩基技术规范》 (JGJ94-2008)

《电力工程地基处理技术规程》 (DL/T5024-2005)

《建筑桩检测技术规范》 (JGJ106-2014)

《混凝土结构工程施工质量验收规范》(GB50204-2015)

《建筑桩基设计规范》 (JGJ 94-2008)

《建筑地基基础工程施工质量验收规范》(GB50202-2013)

《混凝土结构设计规范》 (GB 50010-2010)

《变电站建筑结构设计技术规程》 (DL/T 5457-2012)

《建筑地基基础设计规范》 (GB50007-2011)

12.2. 当桩基施工完成后，应由业主委托有相应资质的检测单位对桩基进行检测，检测单位编制检测大纲报业主、监理部审查。

12.3. 应对桩身混凝土质量和承载力进行检测，测试方法为高、低应变动测法。

（1）桩身混凝土质量检测：低应变动测法，数量不少于总桩数的50%，且不得少于20根；柱下三桩或三桩以下的承台抽检桩数不得少于1根。

（2）桩的竖向承载力检测：高应变动测法，数量不少于总桩数的5%，且不得少于5根。

（3）工程桩检测时，为避免高应变动力检测砸坏工程桩桩顶，需检测的工程桩桩头应按《建筑基桩检测技术规范》的要求提前进行加固处理。

12.4 桩基竖向抗拔承载力检测（静载荷试验法）：检测数量不少于3根，且不应少于总桩数的1%，抽样应有代表性。

12.5 验收：验收前施工单位应向业主（或监理部）提供经工地技术负责人认定合格的自检记录、施工记录及工程桩检测报告，作为合格工程完工后进行竣工验收的主要依据。

13.桩身混凝土强度采用fc'=28MPa，水胶比不大于0.45，氯离子含量不大于0.1%，水泥应满足ACI-318-14表19.3.2.1中暴露类别S1对应水泥（Cementitious materials-Types）的要求，抗渗等级0.8MPa，掺入矿物掺合料。

14.其它未尽事宜请严格按照有关的规程规范、标准图集等的相关内容执行。

15.本工程不同位置的桩如有特殊说明，详见各桩基布置图。

16.本工程初设设计尚未审定，未通过NTDC审核，桩基存在修改的可能，施工前请联系业主与设计确认。

1.Summary

1.1 The site level of ±0.000 is equal to the absolute elevation of193.40 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.

1.2 The foundation design of the project proceeds according to《The geotechnical engineering investigation report(Detail survey) of drawing construction design phase of ±660KV converter station in Lahore in Pakistan Matiari ±660KV HVDC project》(B003701S-G01-01).

Investigation report shows that liquefaction may occur within 12 meters of the engineering site .

1.3 The bored piles in the project are adopt to end bearing friction plies, the bearing stratum at the tip of piles are 2 to 3 or 2 to 4 layers of fine sand.

1.4 Referring to the National building standard design atlas《Reinforcement concrete bored pile》Number of atlas:10SG813;

2.Boring

2.1 During the construction of pile foundation,sediment at bottom of hole should be cleaned after the hole is drilled and the steel cage is installed.After cleaning, the sediment thickness shall not be greater than 100. According to the regulation, the concrete must be poured immediately within half an hour. Otherwise, it should be re cleaned to meet the requirement.

2.2 When the hole reaches the design depth, the orifice shall be protected and shall be checked and accepted according to relevant regulations, and make records properly.

2.3 In the construction process,the pore size of piles shall be checked uniformly and randomly, and the quantity shall be greater than 10% of the total number of piles.

3. Fabrication and installation of steel cage

3.1 The main reinforcement, spiral stirrups and stiffening stirrups of bored piles are welded into reinforcing bars. If welded in sections, the cage shall be welded.Vertical and horizontal reinforced junction shall be welded, welding rod by E43 and E50.

3.2The main reinforcement and stirrup do not hook.A minimum stirrup should be aligned at the bottom of each main reinforcement.The number of welded joints of main reinforcement in the same section shall not be greater than half.

3.3 Concrete blocks or other effective measures are needed on the outside of the reinforcement cage,to ensure that the reinforcing cage main reinforcement protection layer thickness is 75mm

3.4 Proper measures should be taken to prevent twisting and bending when handling the steel cage.Then aim at the hole, lift straight, steady and sink slowly to avoid the collision of hole wall The steel cage shall be fixed to the design position.

3.5 Allowable deviation of steel cage fabrication

Main reinforcement spacing	±10mm	Stirrup spacing	±20mm
Diameter of steel cage	±10mm	Length of steel cage	±100mm

3.6 The joint of sectional reinforcement cage shall be welded(The diameter of the rebar is greater than 20mm), And it should be observed by the current national standard "general technical regulations for steel machinery connections", JGJ 107, code for welding and acceptance of reinforcing steel bars JGJ 18 and code for acceptance of construction quality of concrete structures GB 50204.

3.7 Sstiffener setting is referred to Atlas of (10SG813).

3.8 The outer diameter of the pipe joint should be smaller than the inner diameter of the steel cage, 100mm or more.

4. The concrete filling coefficient should be bigger than1.05.

5. Allowable deviation of construction

5.1 The allowable deviation of pile diameter D is +50(negative deviation is not allowed).

5.2 The allowable deviation of the pile perpendicularity is 1/100.

5.3 The allowable deviation of center position of pile:

(1) The deviation of the single row piles and strip foundation along the vertical axis direction and group piles foundation is D/6 and not greater than 100.But two adjacent piles can not be partial to same direction.

(2) The deviation of the strip pile foundation along the axis direction and the group piles foundation is D/4 and not greater than 150.But two adjacent piles can not be partial to same direction.

(3) The allowable deviation of the center position of single pile under column is D/6 and not greater than100(D is the diameter of the pile).

6. Monitoring should be carried out during construction, including observation of pore water pressure,displacement at pile top,soil displacement, etc.According to the monitoring results, the reasonable rate and sequence of pile formation are controlled. Necessary measures are taken to ensure the quality .

7. After a pile is completed, the foundation pit shall be suspend for some time,The rest time and date excavation shall be determined according to the requirements of relevant specifications and excess pore water pressure dissipation.

8. Pile forming sequence principle:

8.1 Depending on the density of the pile,the piling can be carried out symmetrically from the middle to the two directions,from the center to periphery, from one side to single direction.

8.2 According to the design elevation of the foundation, it should be deep firstly and then shallow.

8.3 According to the specification of pile, it should be big first and then small,first long and short afterwards.

8.4 When one side is adjacent to the building, constructing from adjacent buildings to another direction.

8.5 For different types of pile, such as pipe pile,cast-in-place pile construct with mixing pile and gravel pile , it should be in accordance with the actual situation, consulting the pile construction sequence.

9. Before the excavation of the foundation pit, the construction plan shall be worked out for slope stability, dewatering measures, excavation scheme, soil handling route and pile location.The excavation within the pit shall be carried out in layers, and the elevation difference shall not exceed 1 meters.it shall not be piled on the side , affecting the stability of the pile body and preventing the displacement and inclination.

10. If the geological conditions are not in conformity with the survey data or other unexpected circumstances during the piling process, please contact the Design Institute in time.

11. The connection structure between the pile and the cap refers to the 34 page of atlas (10SG813).The main reinforcement extends into the pile cap with a length of 40d.

12. Quality inspection and acceptance

12.1 In addition to the above provisions, the quality standards shall also be subject to the relevant provisions in the following current norms:

《Design, Manufacture, and Installation of Concrete Piles》 ACI-543R-12

《Technical specification for pile foundations of buildings》(JGJ94-2008)

《Technical specification for ground treatment of power engineering》(DL/T5024-2005)

《Technical specification for inspection of building foundation piles》(JGJ106-2014)

《Standard Test Method for mechanical properties of ordinary concrete》(GB/T50081-2016)

《Code for acceptance of construction quality of building foundation》(GB50202-2013)

《Code for design of concrete structures》 (GB 50010-2010)

《Code for design of building foundation》 (GB50007-2011)

《Code for design of pile foundations of buildings》(JGJ 94-2008)

《Code for acceptance of construction quality of concrete structures》(GB50204-2015)

《Technical specification for design of substation building structures》(DL/T 5457-2012)

12.2. When the construction is completed, the owner shall entrust the testing unit with corresponding qualification to test the pile foundation.

12.3. The concrete quality and bearing capacity should be tested, and the test method should be high and low strain method.

(1) Quality test of pile concrete: low strain method, no less than 50% of the total number of piles, and shall not be less than 20;

(2) Test of vertical bearing capacity of pile:high strain method,the quantity shall be no less than 5% of total number, and no less than 5.

(3) To avoid the smashing of high strain dynamic testing of pile top,the pile shall be reinforced in advance according to the requirements of the technical specification for building foundation test.

12.4 The vertical uplift bearing capacity of the pile fondation detection(static load test method) as follows:testing quantity not less than three root,sample should be representative.

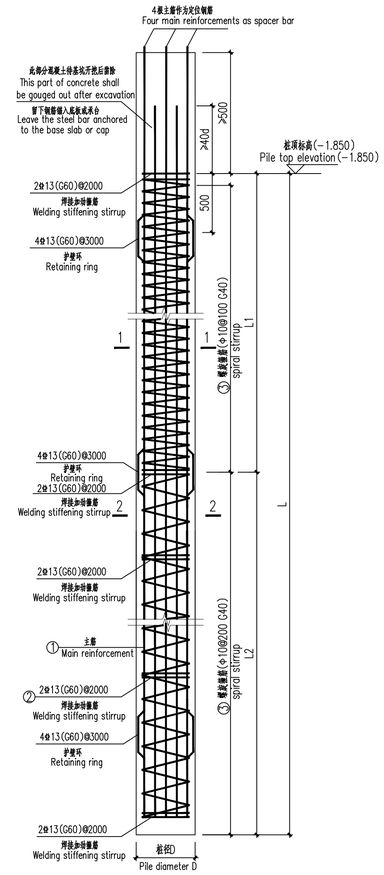
12.5 Acceptance: before acceptance, the construction unit shall provide the owner (or the supervision department) with the record of qualified self inspection records, construction records and engineering pile inspection reports as the main basis for final acceptance.

13.The concrete strengh of the pile body fc' is 28MPa,the water binder ratio is not greater than 0.45,the content of chloride ion is not more than 0.1%,The cement shall meet the requirements of the exposure category S1 in19.3.2.1 in Table ACI-318-14, corresponding to the cement (Cementitious, materials-Types), the impermeability grade is 0.8MPa, and the mineral admixture is added.

14.Other matters not related, please follow the relevant procedures, specifications, standards, Atlas and other related content strictly.

15.If there is any special description of the pile at different locations of the project, referring to the layout of each pile foundation.

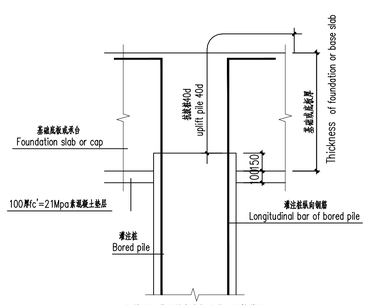
16.The initial design of the project has not yet been approved by NTDC, the pile foundation is possible to amend.Please contact the owner and designer before construction.



工程桩详图

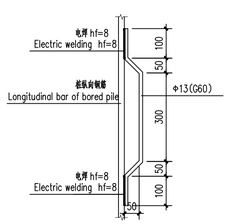
Detail drawing of the pile

BZ01、BZ02



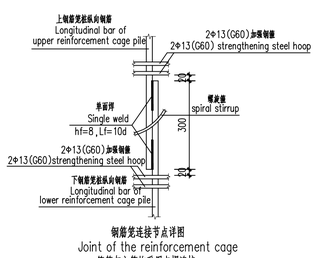
机械成孔灌注桩与底板或基础连接节点图

Joint of the bored pile and bottom plate or foundation



护壁环详图(可用混凝土空心垫块代替)

Retaining ring (Concrete hollow block is available)

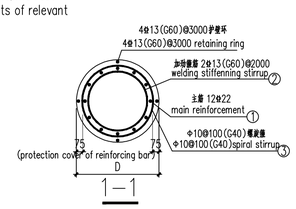


钢筋笼连接节点详图

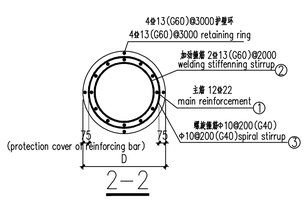
Joint of the reinforcement cage

箍筋与主筋均采用点焊连接

Both the stirrup and the main reinforcement are welded by spot welding



1—1



2—2

桩型编号	桩径D (mm)	L=L1+L2(m)	混凝土强度等级	单桩承载力设计值(kN)			L1	L2	桩身配筋				
				抗压	抗拔	水平			钢筋笼长(m)	主筋①	加劲箍筋②	螺旋箍筋③	
BZ01	600	28	fc'=28Mpa	549.9	504.7	102.2	16	12	L1+L2	28	12Φ22(C60)	2Φ13(C60)@2000	Φ10@100/200(C40)
BZ02	600	28	fc'=28Mpa	629.2	559.9	102.2	16	12	L1+L2	28	12Φ22(C60)	2Φ13(C60)@2000	Φ10@100/200(C40)

No.	diameter D(mm)	L=L1+L2(m)	Concrete strength grade	Design value of bearing capacity of single pile (kN)			L1	L2	Reinforcement of pile body				
				Compressive bearing capacity	Uplift bearing capacity	Horizontal bearing capacity			The length of steel cage(m)	main bar ①	welding stiffenning stirrup ②	spiral stirrup ③	
BZ01	600	28	fc'=28Mpa	549.9	504.7	102.2	16	12	L1+L2	28	12Φ22(G60)	2Φ13(G60)@2000	Φ10@100/200(G40)
BZ02	600	28	fc'=28Mpa	629.2	559.9	102.2	16	12	L1+L2	28	12Φ22(G60)	2Φ13(G60)@2000	Φ10@100/200(G40)

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 6 1 7 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	龙金来	
校核 Checked	王惠平	
审核 Reviewed	吴春晖	
批准 Approved	李和平	

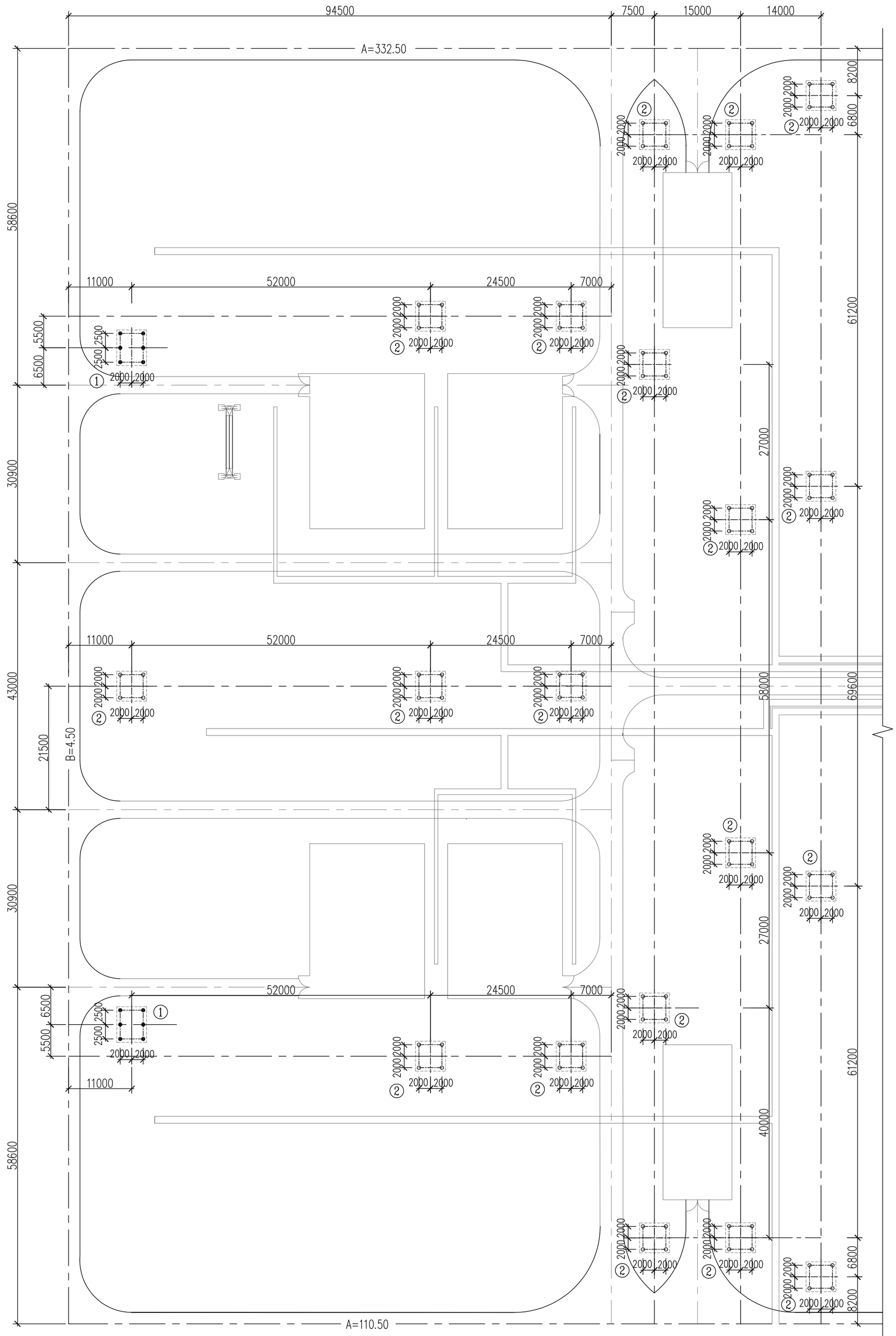
灌注桩设计说明

specification for cast-in-place pile

比例 Scale:

图号 DWG NO.: 30-B003701S-T0617-001

版本 REV.: A



桩位图 layout of piles

说明：

1、场地±0.000相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。

2、本工程基础采用柱下独立承台+灌注桩的形式，桩为Φ600的机械成孔灌注桩，

● 表示工程桩(BZ01)，为Φ600机械成孔灌注桩，桩长28米，桩顶标高为-1.850m，
桩身砼强度fc'=28Mpa，桩端持力层为②3层粉细砂层或者②4层粉细砂层;
单桩承压承载力容许值为549.9kN，抗拉承载力容许值为504.7kN,水平承载力容许值为75kN。共12根。

○ 表示工程桩(BZ02)，为Φ600机械成孔灌注桩，桩长28米，桩顶标高为-1.850m，
桩身砼强度fc'=28Mpa，桩端持力层为②3层粉细砂层或者②4层粉细砂层;
单桩承压承载力容许值为629.2kN，抗拉承载力容许值为559.9kN,水平承载力容许值为75kN。共76根。

3、对施工的要求：
3.1.机械成孔灌注桩施工要求详见30-B003701S-T0617-001。
3.2.施工单位在确定沉桩时，所选用的施工设备，应确保桩位的准确，确保桩身质量的完整，并护好周围道路及环境；施工中发现贯入度异常或与勘察资料土质不符，及时通知设计人员现场研究解决。

4、本项目桩基设计等级为丙级。

5、桩基在施工过程中，除满足本说明的要求外，应严格执行《Design,Manufacture, and Installation of concrete piles》(ACI 543R-12)的相关规定。

6、桩基钢筋防腐应控制混凝土配料中所含可溶氯离子浓度最大含量不超过0.1%控制，测试方法参《ASTM C 1218M》。

7、未注明桩定位均关于轴线对中。

8、本工程初设设计尚未审定，未通过NTDC审核，桩基存在修改的可能，施工前请联系业主与设计确认。

9、由于线路尚未经巴方审查，若极线塔位置调整可能引起避雷线塔位置及桩基方案修改，是否按图施工请中电装备和监理确认。

Specification :

1. The site level of ±0.000 is equal to the absolute elevation of193.40 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.

2、The foundation of this project adopts the form of independent cap and filling pile under column.
The pile is a bored cast-in-place pile with a diameter of 600.
● stands for the engineering pile(BZ01).The length of the pile is 28 meters.The elevation of pile top is -1.850m .
The concrete strength of the pile body fc' is 28MPa.The bearing stratum at the tip of piles are 2 to 3 layers or 2 to 4 layers of fine sand.
The allowable compressive bearing capacity of single pile is 549.9kN. The allowable uplift bearing capacity of single pile is 504.7kN.
The allowable horizontal bearing capacity of single pile is 75kN. The number of the piles is 12.
○ stands for the engineering pile(BZ02).The length of the pile is 28 meters.The elevation of pile top is -1.850m .
The concrete strength of the pile body fc' is 28MPa.The bearing stratum at the tip of piles are 2 to 3 layers or 2 to 4 layers of fine sand.
The allowable compressive bearing capacity of single pile is 629.2kN. The allowable uplift bearing capacity of single pile is 559.9kN.
The allowable horizontal bearing capacity of single pile is 75kN. The number of the piles is 76.

3、Requirements for construction:
3.1.Requirements for construction of mechanical bored piles are detailed in 30-B003701S-T0617-001.
3.2.The construction equipments used by the construction unit in determining the pile shall ensure the accuracy of the pile location, the integrity of the pile quality, and the protection of the surrounding roads and environment;

4、Design class of foundation is class C.

5、In addition to meeting the requirements of the specification, the relevant regulations in 《Design,Manufacture and Installation of concrete piles》(ACI 543R-12) should be strictly enforced. ,

6、The concentration of soluble chloride ion in concrete burden shall be controlled.The maximum content shall not be more than0.1%. Test method refers to "ASTM C 1218M".

7、Undated piles are symmetric on the axis.

8、The initial design of the project has not yet been approved by NTDC, the pile foundation is possible to amend.Please contact the owner and designer before construction.

9、For the transmission line hasn't been checked by Pakistan,the tower position and pile design may be modified by the position adjustment of pole line tower. Please contact Electric Power Equipment and supervisor before construction.

专业 Specialty	会签 Sign	日期 Date
变电一次		

签名 Name	日期 Date
设计 Designed	龙金来
校核 Checked	孔春子
审核 Reviewed	吴俊清
批准 Approved	李松涛
比例 Scale	1:400
图号 DWG NO.	30-B003701S-T0617-002

Revisions

Name	Date	Notes

Reference Drawings

Drwg. No. | Title

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

CONTRACTOR: 国家电网 STATE GRID; 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT & TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站
±660kV Lahore Converter Station

SUBCONTRACTOR: East China Electric Power Design Institute Co.,Ltd.

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

设计阶段 PHASE: 施工图 Detail Design

版本 REV. A

直流防雷线塔桩位图
layout of piles for tower of guard cable